A Low-Cost IIoT Multi-Sensor Platform for Real-Time Electric Motor Fault Detection

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***Abstract*—Unexpected electric motor faults can cause downtime, secondary damage, and increased maintenance cost in industrial systems. This paper presents a low-cost Industrial Internet of Things (IIoT) platform for real-time electric motor condition monitoring and preliminary mechanical fault detection. The proposed system integrates an Arduino Nano 33 IoT, its onboard LSM6DS3 inertial measurement unit, an SHT11 temperature sensor, a shunt-based current sensor, and an A3144 Hall-effect speed sensor. Instead of continuously transmitting raw vibration samples, the embedded node extracts root mean square and crest factor features from three-axis acceleration data over a 20 s monitoring window. Real-time data are published through Message Queuing Telemetry Transport (MQTT) to a local Node-RED dashboard and a Python inference application, while selected measurements are logged to ThingSpeak through Hypertext Transfer Protocol (HTTP). Experimental data were collected under healthy low-speed, healthy high-speed, loose-screw/soft-foot, and shaft-misalignment conditions. Exploratory analysis showed that vibration-based features provided the clearest class separation, whereas temperature and current were less discriminative under the tested load and speed range. A support vector machine with a radial basis function kernel achieved 95.0% test accuracy, compared with 90.6% for logistic regression. The results demonstrate the feasibility of combining edge feature extraction, lightweight IoT communication, and machine learning for low-cost motor health monitoring.**

***Index Terms*—condition monitoring, electric motor fault detection, IIoT, machine learning, MQTT, predictive maintenance, vibration sensing**

I. Introduction

Electric motors are essential components in industrial automation, manufacturing, pumping, ventilation, robotics, and transportation systems. Their faults can lead to unplanned downtime, secondary damage, reduced productivity, and increased maintenance cost. Conventional reactive maintenance repairs equipment after failure, while time-based preventive maintenance can result in unnecessary service actions. Condition-based and predictive maintenance aim to overcome these limitations by continuously observing equipment health and detecting abnormal behaviour before severe failure occurs [1]–[3].

Recent developments in low-cost sensors, wireless communication, and edge computing have made Industrial Internet of Things (IIoT) approaches attractive for machine health monitoring. In an IIoT architecture, sensing nodes can measure vibration, speed, current, and temperature, while local or cloud-based software performs visualisation, data logging, and fault classification [4]–[6]. Data-driven predictive-maintenance surveys also identify data acquisition, feature extraction, and machine-learning model selection as key steps for practical industrial fault diagnosis [7]. For rotating machines, vibration analysis is particularly important because mechanical faults such as looseness, imbalance, bearing defects, and misalignment often appear as changes in vibration amplitude, impulsiveness, or frequency content [8]–[11].

However, many research-grade monitoring systems require dedicated data acquisition hardware, external accelerometers, or continuous transmission of high-rate raw data. These requirements can increase cost and bandwidth consumption, particularly when several machines are monitored. For preliminary fault screening and teaching-laboratory deployment, a compact system that extracts informative features at the sensing node is beneficial.

This paper presents a proof-of-concept IIoT motor health monitoring platform based on low-cost hardware and lightweight communication. The main contributions are as follows:

- an integrated multi-sensor node for vibration, temperature, current, and rotational speed monitoring of a DC motor;
- an edge feature extraction approach that calculates root mean square (RMS) and crest factor (CF) features from three-axis acceleration data during each 20 s monitoring window;
- a combined local and cloud communication architecture using MQTT for real-time monitoring and HTTP/ThingSpeak for long-term logging; and
- a preliminary machine learning evaluation comparing support vector machine (SVM) and logistic regression classifiers for healthy, soft-foot/loose-screw, and shaft-misalignment detection.

II. Related Work

Low-cost IoT and IIoT platforms for motor monitoring have recently explored different sensor combinations, communication layers, and learning algorithms. Prasad et al. developed an ESP32-based system for induction motor stator-winding fault monitoring using voltage, current, temperature, and humidity measurements with cloud-based visualisation [12]. Truong et

al. proposed a portable IoT and artificial-intelligence device that uses vibration data for real-time electric motor condition monitoring and smartphone alerts [13]. Widikda and Frayudha developed an IoT-based predictive maintenance system for water-treatment motor agitators using multi-sensor monitoring and machine learning [14]. Kumar et al. investigated predictive maintenance of electric motors using RPM data and machine learning for remaining useful life estimation [17], while Pandey et al. reported an IoT-based motor predictive-maintenance approach using vibration, current, and temperature data [18].

Edge processing is also becoming important because continuous transmission of raw vibration data can increase communication load and energy consumption. Romanssini et al. demonstrated an IoT vibration sensor node with local edge computing for industrial motor monitoring [15], while Aragones et al. presented an autonomous NB-IoT vibration-monitoring node with on-device computation for industrial environments [16]. Previous work by the authors has investigated sensor-based BLDC parameter measurement under different operating conditions and hybrid FPGA-MCU hardware for real-time sensor fusion [19], [20]. Complementary BLDC motor design studies have also shown that torque ripple and vibration reduction remain important in compact motor systems [21]. The present platform extends these directions toward a compact Arduino Nano 33 IoT implementation that transmits only RMS and CF vibration features over MQTT for supervised classification of mechanical faults on a low-cost teaching-scale DC motor testbed.

III. System Architecture

The proposed monitoring system consists of a sensor node, local communication and dashboard layer, cloud logging layer, and machine learning inference layer. The hardware and wiring arrangement are shown in Fig. 1, while the communication architecture is shown in Fig. 2.

A. Sensing Node

The sensing node is built around an Arduino Nano 33 IoT, which includes a SAMD21 microcontroller, Wi-Fi connectivity, and an onboard LSM6DS3 inertial measurement unit (IMU). The IMU measures acceleration along the x, y, and z axes for vibration monitoring. The SHT11 digital temperature sensor measures motor surface temperature. A shunt resistor is used for current measurement, and an A3144 Hall-effect sensor measures rotational speed by detecting magnetic pulses from a magnet attached to the rotating shaft.

Table I summarises the main hardware components and their roles. The design intentionally uses low-cost and readily available components. In particular, the onboard IMU eliminates the need for a separate accelerometer module, and the shunt resistor provides a simple current-sensing option for the tested DC motor.

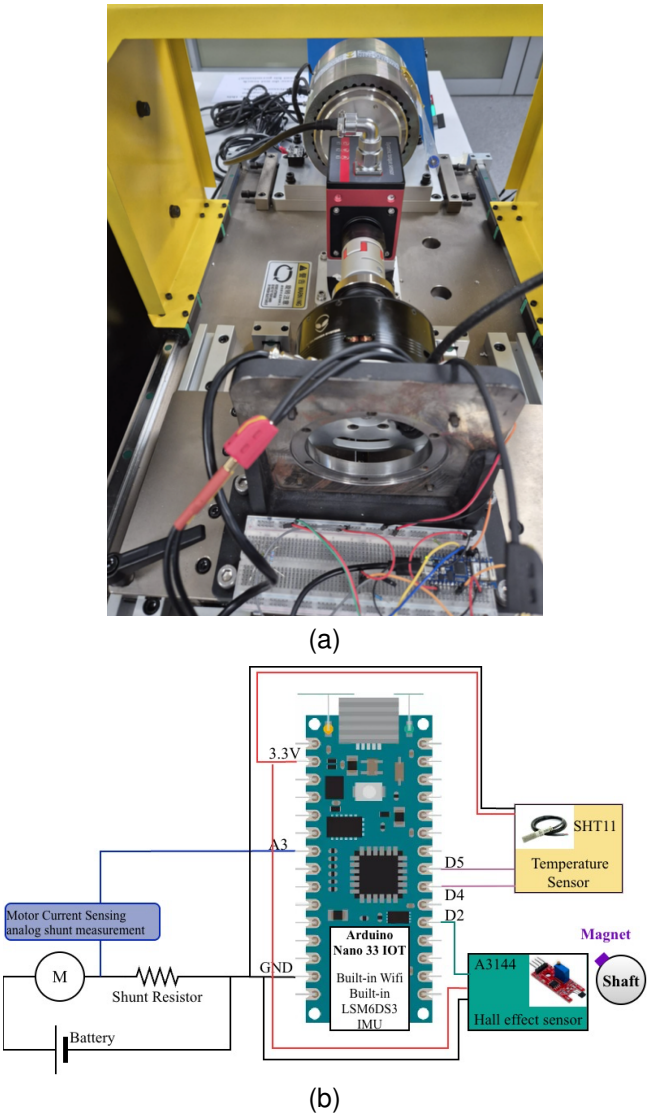


Fig. 1. Implemented hardware and wiring diagram of the low-cost electric motor health monitoring node. (a) Testbench with the whole setup. (b) Wiring diagram of the setup.

TABLE I
Main Hardware Components of the Monitoring Node

Component	Interface	Measured variable
Arduino Nano 33 IoT	Wi-Fi, digital, analog, I^2C	Edge processing and communication
LSM6DS3 IMU	I^2C, onboard	Three-axis vibration
SHT11	Two-wire digital	Motor surface temperature
Shunt resistor	Analog input	Motor current
A3144 Hall sensor	Digital input	Rotational speed

B. Communication and Dashboards

Two communication mechanisms are implemented. MQTT is used for low-latency local data streaming from the Arduino node to a Mosquitto broker, Node-RED dashboard, and Python-based fault prediction application. MQTT was

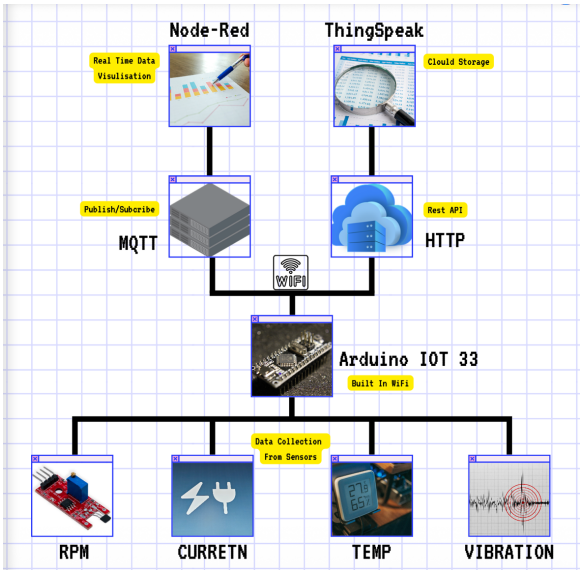


Fig. 2. Communication architecture using MQTT for local real-time monitoring and HTTP/ThingSpeak for cloud data logging.

selected because its publish-subscribe structure is lightweight and suitable for real-time IoT data exchange. HTTP is used for selected cloud uploads to ThingSpeak, enabling remote visualisation and historical data storage.

IV. Edge Feature Extraction

Continuous transmission of raw acceleration samples can increase communication load. Therefore, the proposed system extracts vibration features at the edge and transmits compact feature values. During each 20 s monitoring window, acceleration samples are collected from the x, y, and z axes. For each axis, the RMS value and CF are calculated. The implementation reports one feature vector per monitoring window, consisting of six vibration features and the simultaneously measured auxiliary quantities.

For a sequence of N acceleration samples x_i, the RMS value is calculated as

$$x_{\mathrm{RMS}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}. \tag{1}$$

The RMS value represents the overall vibration energy level. The CF is defined as

$$\mathrm{CF} = \frac{\max(|x_i|)}{x_{\mathrm{RMS}}}. \tag{2}$$

The CF is sensitive to impulsive vibration and abnormal mechanical events. The same calculations are applied independently to all three axes, resulting in six vibration features: RMS-x, RMS-y, RMS-z, CF-x, CF-y, and CF-z.

Motor speed is calculated using the Hall-effect sensor pulse interval. During experiments, false pulses were observed due to sensor noise and switching effects. A simple debouncing rule was therefore implemented: pulses arriving within 50 ms of the previous accepted pulse were rejected. This improved speed measurement stability for the tested speed range.

V. Experimental Dataset

Experimental data were collected using a DC motor under four operating conditions. The first condition was healthy operation at low speed, approximately 500–600 r/min. The second condition was healthy operation at high speed, approximately 1030 r/min. The third condition introduced a loose-screw/soft-foot fault at approximately 1030 r/min. The fourth condition introduced shaft misalignment at approximately 1030 r/min. Table II summarises the operating conditions and the main experimental observations, while Fig. 3 shows representative dashboard outputs for the high-speed healthy and shaft-misalignment cases. During the high-speed tests, the motor current remained nearly constant at approximately 3.66–3.67 A and the motor surface temperature remained within approximately 24.9–25.6°C. This behaviour supports the subsequent feature-selection decision, because the auxiliary current and temperature channels did not provide strong separation between the tested high-speed fault classes.

The collected data were cleaned, labelled, merged, and shuffled before machine learning. Although data were collected under four operating conditions, the classifier used three health labels: healthy, soft-foot/loose-screw, and shaft-misalignment. The low-speed and high-speed healthy measurements were therefore treated as healthy examples. The final feature matrix used for classification contained the six vibration features. Temperature, current, and speed were retained for monitoring but were not used by the final classifier because exploratory analysis showed limited class discrimination under the tested load and speed range. Temperature was measured on the motor casing rather than directly at a bearing housing, and the datasets were collected sequentially, so motor warm-up could influence the temperature trend. Current also showed limited variation because the experiments were conducted at similar load and speed conditions.

VI. Machine Learning Methodology

Exploratory data analysis was used to assess feature separability before model training. Boxplots, correlation analysis, pairplots, and principal component analysis (PCA) indicated that the vibration features provided useful separation between the healthy, soft-foot, and misalignment classes. Example feature distributions are shown in Fig. 4, and the two-dimensional PCA projection of the standardised vibration features is shown in Fig. 5.

The labelled dataset was split into training and test subsets using a stratified 80/20 split to preserve class proportions. Features were standardised using a standard scaler fitted only on the training data and then applied to the test data. Two classifiers were evaluated: an SVM with a radial basis function (RBF) kernel and logistic regression. Hyperparameters were tuned using grid search with five-fold cross-validation. Model performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrices.

TABLE II
EXPERIMENTAL OPERATING CONDITIONS AND OBSERVED SENSOR RESPONSES

Operating condition	Speed	Current	Temperature	Main vibration observation
Healthy, low speed	500–600 r/min	≈3.66 A	21–24°C	Baseline healthy response; crest factors remained low and stable, with no clear fault signature.
Healthy, high speed	≈1030 r/min	≈3.67 A	≈25°C	RMS-y was about 0.0085 g, and crest factors were typically about 1.4–1.7 after speed stabilisation.
Loose-screw/soft-foot	≈1030 r/min	≈3.66 A	≈24.9°C	RMS-y increased to about 0.0097–0.010 g; periodic mechanical impacts were visible in the RMS and CF features.
Shaft misalignment	≈1030 r/min	≈3.66 A	≈25.6°C	RMS-y showed a large transient of about 0.076–0.084 g at fault initiation and settled near 0.011–0.012 g; CF-y exceeded 15.

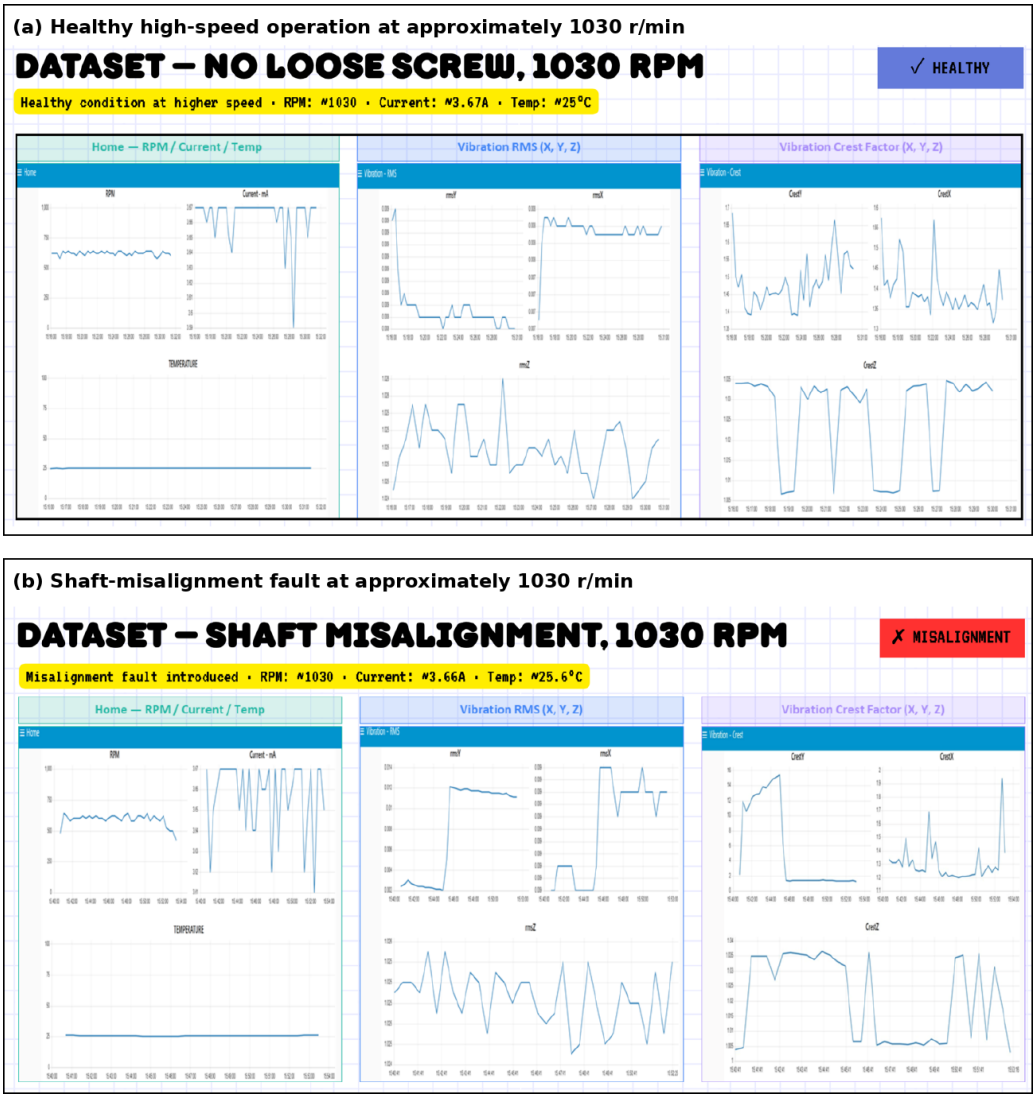


Fig. 3. Representative experimental dashboard outputs for two high-speed test cases: (a) healthy operation and (b) shaft-misalignment fault, both at approximately 1030 r/min. The comparison shows that current and surface temperature remained similar, while the vibration RMS and crest-factor responses changed more clearly under misalignment.

VII. RESULTS AND DISCUSSION

Fig. 6 shows the test-set confusion matrices of the two evaluated classifiers. The test set contained 201 feature vectors, comprising 67 healthy, 69 misalignment, and 65 soft-foot/loose-screw examples. The SVM with RBF kernel correctly classified 191 of these 201 examples, achieving 95.0% test accuracy. Logistic regression correctly classified 182 examples, corresponding to 90.6% test accuracy. The improved

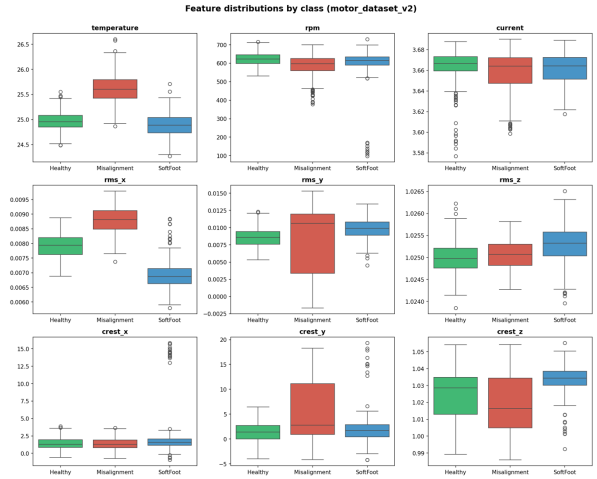


Fig. 4. Boxplots of the measured features showing class-dependent variation in vibration features.

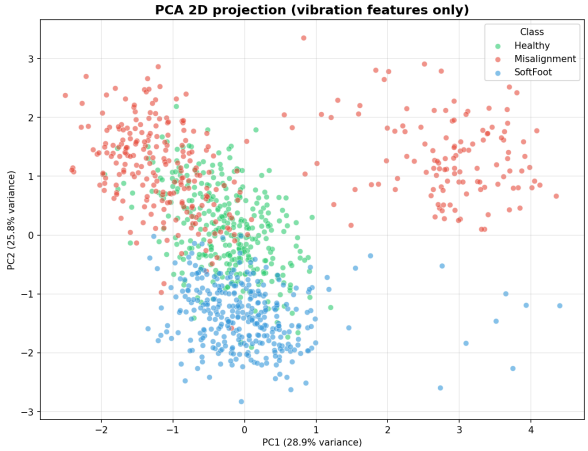


Fig. 5. PCA projection of the six standardised vibration features. The first two principal components explain 54.7% of the variance.

TABLE III
TEST-SET CLASSIFICATION PERFORMANCE

Model	Acc.	Macro F1	Healthy	Misal.	Soft-foot
			Recall (%)		
SVM-RBF	95.0	0.95	97.0	95.7	92.3
Logistic regression	90.6	0.91	94.0	85.5	92.3

SVM performance suggests that the class boundaries are not fully linearly separable in the standardised feature space, which is consistent with the partial class overlap visible in the two-dimensional PCA projection. The macro-averaged F1-score of the SVM was approximately 0.95, indicating balanced performance across the three target classes in the available test dataset. Given the limited size of the test set, these figures should be interpreted as indicative rather than as precise estimates of generalisation performance.

The classification results in Table III suggest that the vibration features are informative for the tested mechanical

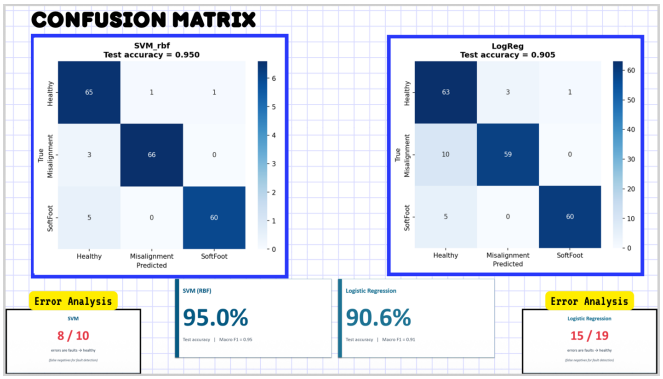


Fig. 6. Confusion matrices and test accuracies for SVM with RBF kernel and logistic regression classifiers.

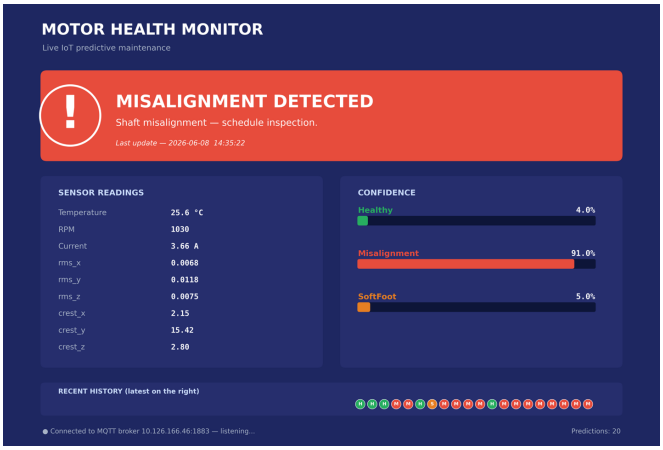


Fig. 7. Real-time monitoring dashboard showing the detected motor condition, confidence values, live sensor readings, and prediction history.

faults. The most common error for both classifiers was a faulty condition being classified as healthy, which is important from a fault-screening perspective. The SVM reduced these false-healthy cases from 15 to eight, indicating a lower missed-fault rate than logistic regression. The experimental observations in Table II provide the physical explanation for this behaviour: the loose-screw/soft-foot condition increased RMS-y relative to the healthy baseline, while the shaft-misalignment condition produced larger transients and high CF values. In contrast, current and temperature were not sufficiently discriminative under the limited experimental conditions. This does not imply that these variables are generally unimportant for motor diagnostics; rather, their usefulness depends on sensor placement, motor loading, fault severity, and the type of fault being studied.

The trained SVM model was exported and used in a Python real-time prediction application. The application subscribes to MQTT vibration-feature topics, applies the same preprocessing strategy, and displays the predicted motor state, confidence scores, live sensor readings, prediction history, and MQTT connection status. An example of the real-time dashboard is shown in Fig. 7.

The main advantages of the proposed platform are its low cost, compact hardware, simple communication stack, and ability to perform feature extraction at the edge. These characteristics make the system suitable for educational laboratories, early prototyping, and preliminary condition-screening studies. Nevertheless, several limitations should be addressed before industrial deployment. First, the IMU was mounted through the development board and breadboard arrangement, which can reduce vibration coupling compared with direct mounting on the motor or bearing housing. Second, the dataset includes a limited number of operating conditions and two intentionally introduced fault types. Third, the tests were performed over a limited speed and load range. Future datasets should include more machines, repeated trials, different speeds and loads, direct accelerometer mounting, and additional faults such as bearing defects, imbalance, rotor looseness, and electrical faults. Extending the current shunt-based sensing to an isolated current-transformer arrangement would also allow the platform to be applied to AC motors.

VIII. Conclusion

This paper presented a low-cost IIoT-based multi-sensor platform for electric motor condition monitoring and preliminary mechanical fault detection. The system uses an Arduino Nano 33 IoT with vibration, temperature, current, and RPM sensing, and it combines edge feature extraction with MQTT-based local monitoring and HTTP-based cloud logging. Experimental observations showed that the high-speed fault cases produced distinct vibration signatures while current and surface temperature varied only slightly under the tested conditions. Six vibration features, consisting of RMS and crest factor values from three acceleration axes, were therefore used to classify healthy, soft-foot/loose-screw, and shaft-misalignment conditions. The SVM with RBF kernel achieved 95.0% test accuracy and outperformed logistic regression on the available dataset. Future work will focus on direct IMU mounting, larger and more diverse datasets, additional fault classes, and deployment of the classifier directly on the embedded node using TinyML.

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